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RESEARCH ARTICLE

EmpatheticExchanges: Toward Understanding the Cues for Empathy in Dyadic Conversations

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ABSTRACT This article presents a new research task and dataset to understand empathetic responses in conversations between two people. We formulate the empathy prediction problem as a way to model how people with different empathetic abilities react to a dialog by another person. Our novel approach focuses on dyadic conversations within positive and negative emotional contexts. To tackle the proposed problem, we introduce a new dataset, EmpatheticExchanges, presenting the largest sample size of manually annotated exchanges. Besides, EmpatheticExchanges includes empathy features validated by state-of-the-art research in affective computing, such as VAD vector values. Finally, we provide a baseline pattern-based model with a Closeness Evaluation measure score of 0.597 and a weighted F1 score of 0.452 using a 3-point scale. This model is explainable and performs similarly to popular techniques such as deep learning. Our results validate the quality of our dataset and the use of a pattern-based approach for empathy classification.

INDEX TERMS Affective computing, emotion classification, empathy, empathy detection, pattern classification, text-classification.

I. INTRODUCTION

Empathy is one of the most important qualities of human beings. It can be conceptualized as a psychosocial phenomenon that allows humans to understand and share the feelings of others, or the ability to feel what another person is feeling [1], [2]. It allows for connection and understanding, while also contributing to our ability to act together towards a shared goal [3]. Likewise, it is an important part of communication [4]. In this article, we contribute with a novel approach to predict empathy in communication exchanges. We built upon previous research [4], [5], [6] and expanded its scope toward empathy outside negative emotional contexts.

Although there is no universally accepted empathy definition, we define empathy in this article based on the work by Clark et al. [7] and Cuff et al. [8]. Their definition supports three dimensions of empathy: Cognitive, Affective, and Behavioral. Cognitive empathy is to understand the feelings and emotions of another, Affective empathy means to share and experience those feelings and emotions, and Behavioral empathy is to act in a manner that demonstrates cognitive and affective empathy. We follow this scheme as the behavioral dimension of empathy allows us to explore internal psychological processes by analyzing behavior. Additionally, this definition is versatile enough to address nuances about interactions between two people.

Previous work has been done in empathy detection. As such, we aim to overcome the limitations present in the state-of-the-art. Particularly, regarding datasets. Empathy classification has been applied for responses to news stories [6], [9], mental health support [1], [4], [5], [10], and empathetic utterances [11]. These datasets present certain biases and downsides such as an overt focus on negative situations, disregarding the context of a conversation, and having few data points. As such, we present the *EmpatheticExchanges* dataset. Our novel dataset is an enhanced sample of

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the EmpatheticDialogues (ED) dataset by Rashkin et al. [12]. Consequently, it inherits advantages such as emotional diversity, synchronous communication, and topic variety. Besides, it is enhanced with human-annotated labels and tailored for the empathy detection task by formatting conversations as exchanges with contributions from two participants.

The ED dataset has been extensively used for the task of empathetic text generation [13]. Nevertheless, it has seldom been used for predicting empathy [11]. ED consists of 25k personal dialogues grounded in 32 emotions. In addition, each presents self-report labels for empathy, fluency, and relevance. This makes it a suitable candidate for empathy classification. However, previous work has shown that the labels are highly biased and unreliable [11]. Furthermore, the dataset format makes it difficult to apply in empathy detection. We approached the creation of our dataset with two main objectives: ensuring high-quality manually annotated empathy labels and enhancing the features from ED to permit explainable classification of empathy.

We set up our dataset format in exchanges between a *speaker* and *listener* role. This is one of the biggest changes between EmpatheticExchanges and ED. It was necessary to address that empathy requires two or more people to be present [1]. Treating the dataset as a series of exchanges puts it on par with other state-of-the-art datasets while surpassing their limitations. In addition, it allows us to capture the context of empathetic interactions from both participants.

Another advantage of our dataset is the use of a unified empathy metric. Our metric is based on human evaluation and measures behavioral empathy in text-based interactions. In addition to our labels, we enhance the dataset in terms of features. To do this, we leverage *empathy components* in the dataset, which are subsequently used in our empathy detection framework for explainable classification [3]. These are concepts related to empathy such as cues, emotional states, feelings, or actions. With these cues as features, we can create explainable empathy predictors [5], [11].

Using the empathy components and our human-annotated labels, we applied an explainable AI (XAI) classifier for the task of empathy classification. We used PBC4cip [14], a contrast-pattern classification algorithm for class imbalance problems. The results show that using empathy components combined with an explainable classifier yields comparable results to a state-of-the-art BERT [15] classification algorithm. As such, we present a new benchmark using our novel dataset.

Our contributions are as follows:

1) We present EmpatheticExchanges, a novel dataset of empathetic exchanges grounded in emotional situations. This dataset has as a main feature a novel metric for behavioral empathy. Additionally, it contains numerous values for empathy components about the exchange. Our empathy metric is manually annotated and supported by psychological research. To our knowledge, this is the largest dataset with these characteristics.
2) We carry out empathy detection using traditional machine learning, explainable, and state-of-the-art algorithms. These algorithms predict the perceived levels of behavioral empathy given in text responses. We present a benchmark for this task.
3) We detail the contribution of empathy components to the overall perception of empathy in conversations using an ablation study. We show which elements are more important than others, and which contribute negatively to performance.

Our dataset can benefit the empathetic classification and text generation research community. Additionally, we expect that this research will impact other fields. For example, gaining insight on the text-based interactions that are increasingly more common thanks to the internet [1]. Likewise, our work can help shed light on the specific characteristics of empathetic communication. Empathy is a complex subject with characteristics that are difficult to pinpoint. Therefore, our research findings regarding two-person (dyadic) conversations can be useful for psychological research.

Our article is presented as follows: Section II explores the related work and background. Section III presents our methodology for building the dataset. Section IV shows our empathy classification approach. Section V describes our results for empathy detection and the ablation study for empathy components. The implications of our research and a discussion of our experimental results are found in Section VI. Finally, we conclude our study and suggest the directions for future research in this field in Section VII.

II. RELATED WORK

Empathy detection in text is a task that presents many challenges. Researchers have contended with the different definitions of empathy, the complexity of the subject, and the subjective nature of empathy measurements [4]. Nevertheless, some progress has been achieved in incorporating empathy in affective computing. This section will explore the most relevant developments related to our research.

A. DATASETS

Developing a dataset for empathy classification depends heavily on the empathy metric chosen for this task. Empathy metrics allow empathy prediction to be modeled as a multi-label, multi-class, or binary classification problem [4]. Therefore, the datasets are built around the empathy metric and a specific empathy definition. We provide a summary of public empathy detection datasets in text interactions between two people (dyadic interactions) in Table 1.

Empathetic concern is a popular metric in empathy classification. It is defined as "feeling for others" by Batson [16]. Therefore, it is often just labeled as *empathy* in affective computing research [6], [9], [17], [18]. This metric was used

to build two datasets for empathy classification. The dataset used in WASSA 2022 for the *empathy and distress prediction* task was compiled by Buechel et al. [9]. In addition, Empathetic Conversations was presented by Omitaomu et al. [6]. Both datasets are created for empathy prediction based on essays written in response to news stories and feature a 7-point scale for empathy. Batson's definition of empathy links empathetic concern and personal distress. Therefore, these datasets and their classification tasks tackle empathy in negative contexts. This is a limitation as they overlook empathy in positive scenarios (positive empathy) [19].

Hosseini and Caragea propose a different empathy metric in their two datasets [1], [10]. They present a model in which seeking and providing empathy are labeled alongside empathy components like sentiment and emotion. Their datasets are built upon the posts from cancer support online communities. Posts present a binary label for *seeking-empathy* and *providing-empathy*. This work shows the importance of considering empathy as the result of an interaction between two people. The results from experiments using the dataset show that binary classifiers can identify empathetic interactions in the context of cancer support using distilled knowledge from related sources. Nevertheless, further work must be carried out to adapt this strategy to other domains.

One of the most influential developments in empathy detection was the work by Sharma et al. [5]. They focus on empathy in online mental health support. Their approach is to use a corpus of text exchanges between a person seeking mental health support on online forums and the responses to those utterances. They compiled data from Reddit and another mental health support forum. Their empathy metric is given by the EPITOME framework, which separates expressed empathy into three communication mechanisms: Emotional Reactions, Interpretations, and Explorations. Alongside a 3-point scale that measures the level of each communication mechanism, they include *rationales*. Rationales are highlighted parts of sentences that formed the reasoning behind the annotation. Their datasets were subsequently used for multi-label empathy classification, in which their proposed model outperformed baseline classifiers. This is an impactful development as it has been used for automatic empathy prediction [4] and empathetic text generation [20], [21], [22], [23].

Montiel-Vázquez et al. [11] modified a subset of EmpatheticDialogues and added human evaluation labels. Their empathy metric is a 5-point scale that describes the empathetic nature of the interaction. They applied an empathetic ability test for each annotator to ensure that the labels were accurate and not arbitrary. Subsequently, they used the labels to predict empathy per utterance. This work is the most similar to ours.

Nevertheless, Montiel-Vázquez et al. use features that make their results difficult to interpret. In addition, unlike Sharma et al., they do not consider empathy part of an interaction and predict the empathy in each utterance separately. Therefore, each utterance presents no context. Furthermore, the data points for this subset of EmpatheticDialogues are few, with only 400 conversations. Their classification results show modifying EmpatheticDialogues is a viable strategy for detecting empathy, but future work should address their limitations.

There are additional datasets for empathy detection that use a domain different from text [24], [25], [26], [27], [28], address more than two people [29], or are not public [30], [31], [32], [33], [34], [35], [36]. However, we focus on empathy detection in dyadic text interactions that can be accessed by the public. We leave different domains to be addressed by future research.

Our dataset combines and adapts features from previous research. We provide multiple-point scales for empathy based on psychological research [11], [17], [37]. Furthermore, we address the direction of empathy [1], [10] using an exchange format [5]. Additionally, our dataset contains multiple empathy components obtained from empathy detection datasets and empathy generation algorithms to incorporate additional knowledge into empathy modeling. Finally, our dataset is the first and largest to provide metrics for positive and negative behavioral empathy in synchronous interactions.

B. LEXICONS

To our knowledge, the only lexicon focused on empathy so far was proposed by Sedoc et al. [18] in 2020. The empathy lexicon was based on Batson's empathetic concern. They used a mixed-level feed-forward neural network to develop word ratings from document-level ratings. In this manner, they obtain a set of informative words and weights for empathetic concern and personal distress that circumvents the difficulty of manually rating each word.

Lexicons continue to be useful even for complex psychological constructs like empathy [18]. This is because they can provide robust results with a relatively simple structure. Therefore, we applied an emotion lexicon [39] for our feature extraction.

C. AUTOMATIC EMPATHY CLASSIFICATION

Amjad et al. [4] proposed the EMP-EVAL empathy automatic evaluation method. Their objective was to avoid human annotation. This classification framework is based on EPITOME by Sharma et al. and includes additional dialogue acts not considered by previous research. Amjad et al. predict emotional (affective) and cognitive empathy factors from a response pair. Furthermore, they identify the dialogue emotion. These factors are used to obtain an overall score of empathy using a weighted sum. This methodology shows an approach for obtaining an empathy score based on empathy components with a high correlation to human annotation.

Amjad et al. showed that using empathy components can increase classification performance. We based our feature enhancement on this principle. We hypothesized that adding

TABLE 1. Comparison table for empathy detection datasets. EmpConv: Empathic Conversations, EC: Empathetic Conversations, Cog&Aff: Cognitive and Affective, self-disc: self-disclosure, Taxo: Taxonomy, 3-mech: three mechanisms., DA: Dialogue Acts.

Database	Characteristics	Empathy	Scale	Components
NewsEmp [9]	News domain, Essays, 1,860 samples	Concern	7-point	Distress
EPITOME [5]	Health domain, Asynchronous exchanges, 10,143 samples	3-mech.	3-point	EPITOME scores, Rationales
IEMPATHIZE [1]	Health domain, Forum messages, 5,007 samples	Perceived	Binary	Sentiment
TweetEmp [10]	Health domain, Tweets, 3,000 samples	Perceived	Binary	Emotion, Polarity
WASSA 2022 [17]	News domain, Essays, 2,555 samples	Concern	7-point	Emotion, Distress, Personality, IRI
EmpConv [6]	News domain,Conversations & essays, 500 samples	Concern	7-point	Emotion, self-dis, Polarity, DA
EC [11]	Emotional Situations, utterances, 1,713 samples	Cog&Aff	5-point	Emotion, Sentiment, Intent, Taxo
AcnEmpathize [38]	Health domain, Forum messages, 12,212 samples	Expressed	Binary	User metadata
Empathetic Exchanges	Emotional situations, Synchronous exchanges, 4,949 samples	Behavioral	3 & 5-point	Emotion, Sentiment, Intent, EPITOME scores, VAD vectors, Mimicry

relevant empathy components like those on the EPITOME framework would have the same effect on our classifier.

D. EMPATHETIC TEXT GENERATION

Another task that must be mentioned is the generation of empathetic dialogue. Pursuing more empathetic chatbots has shown multiple approaches with varied methodologies. Raamkumar and Yang [13] published a survey on empathetic conversational systems. Their observations show performance improvement when incorporating emotion, emotion-cause, sentiment, and other empathy components in the text generation pipeline. As such, newer algorithms have pursued the same avenue since the publication of that survey [23], [40]. We use the findings in this research area to select effective empathy-related concepts for our dataset aimed at classification. We observed an increase in performance for classification consistent with the same trend in empathetic text generation.

E. PATTERN-BASED CLASSIFICATION

Pattern-based classification refers to a type of explainable classification algorithm that has been previously used for empathy classification [11]. Patterns are expressions that describe a subset of the data [41] or regularities observed in a sample subset [42]. These algorithms leverage patterns in the data as new features to label instances. Their main advantage is that they are interpretable unlike the ''black box'' method of deep-learning algorithms [43].

Contrast patterns appear significantly more often in a class than others and they can be used to classify instances more accurately. As such, Loyola-Gonzalez et al. [14] developed the contrast pattern-based classifier for class imbalance problems (PBC4cip). This algorithm uses weights to deal with class imbalance. Therefore, it ensures that the original dataset is not modified to deal with this issue. This has the added benefit that patterns found by this algorithm accurately represent the dataset.

PBC4cip works by first performing pattern mining using a forest-based method. Then, it computes weights for each of the classes in the dataset based on the number of instances in the class, the total instances in the training dataset, and the frequency of patterns that cover each of the classes. Specifics details can be found in the original paper [14]. Our implementation uses 100 trees for pattern mining.

We needed to obtain a classification approach that would work as a benchmark using EmpatheticExchanges. However, we were also interested in determining the effect of different empathy components on classification performance. PBC4cip is an appropriate selection for these tasks because it overcomes data imbalance and outputs interpretable patterns. Therefore, we used PBC4cip to obtain a classification benchmark in our dataset. We compare the results between this classifier, classical machine learning algorithms, and a state-of-the-art approach based on BERT [15]. With this, we explore the feasibility of pattern-based algorithms in contrast to the current paradigm. Our main contribution using PBC4cip is to obtain a benchmark for empathy classification using explainable classifiers and discover patterns that link empathy performance using components adjacent to the concept.

III. EMPATHETICEXCHANGES

Our dataset presents dyadic conversations grounded on emotional situations. These conversations are separated into pairs of utterances with a *speaker* role and a *listener* role. This structure is what we call *exchanges*. The speaker's role presents an utterance first. Then, the listener role responds. We name our dataset *EmpatheticExchanges* due to the importance of this format for capturing empathy in interactions between both roles.

EmpatheticExchanges comprises around $5k$ individual exchanges and aims to label the levels of behavioral empathy displayed by the listener role. We provide our dataset with 3-point and 5-point labels that correspond to the behavioral empathy of the listener. Additionally, each exchange includes features related to empathy components for both roles (speaker and listener). Our features allow a classifier to use information derived from text that offers a deeper additional information besides text.

In summary, we provide a novel dataset for classification tasks containing a new behavioral empathy metric and numerous empathy components. We make this dataset public

alongside its source code for future research.[1] To our knowledge, EmpatheticExchanges is the largest dataset of manually labeled conversations with these characteristics. We provide a graphical representation for the process of creating EmpatheticExchanges in Figure 1.

A. EMPATHY METRIC

We identified that scales with multiple points were more prevalent in the state-of-the-art than binary ones. This is due to the possibility of capturing more nuance. Likewise, previous metrics are substantiated using the Likert scale, a robust psychological tool for measuring affective variables [44].

Our first approach for an empathy metric was the 5-level empathy score (5-ES). We followed the same scale taken by Rashkin et al. [12] and Montiel-Vazquez et al. [11]. This scale has previously been proven useful for measuring empathy and can be easily understood by annotators. Therefore, we can use this metric to obtain the perceived behavioral empathy in an interaction.

Our second metric for behavioral empathy using 3 levels is called 3-ES. This is based on the definition of empathy as a multi-dimensional [7], [8] and hierarchical [45], [46], [47] construct. The levels were delineated using the following criteria:

1) **Little to no empathy**: Presenting up to the most basic level of empathy (mimicry) [48].
2) **Somewhat empathetic**: Displaying cognitive or affective empathy, but not both.
3) **Empathetic**: Behavior that corresponds to experiencing cognitive and affective empathy. A general description of behavioral empathy [7], [47].

We developed the 3-ES since we identified several disadvantages with its 5-level counterpart (5-ES). Firstly, it is difficult to discern between something being ''very slightly empathetic'' and ''not empathetic.'' Likewise, between ''Empathetic'' and ''Very much empathetic.'' These descriptions make it difficult for annotators to create distinct classes for behavioral empathy in responses. Furthermore, these metrics do not correspond to empathy characteristics in psychological literature. Given this, we decided to build a narrower metric based on existing hierarchical models of empathy [45], [46], [47], [48]. In this manner, we obtain a more robust metric that permits nuance.

Both the 5-ES and 3-ES metrics measure the behavioral empathy of the listener. Therefore, both are part of our EmpatheticExchanges framework (EEF) for measuring behavioral empathy. We design level 1 of 3-ES to be equivalent to a score of 1 or 2 in 5-ES. Likewise, a score of 3 will correspond to 4 or 5 in 5-ES. Therefore, 3-ES can be obtained using 5-ES. This makes 3-ES a very versatile metric, as it can be obtained by applying a 5-point Likert scale, as in previous research [11]. However, it presents distinctive characteristics and better backing for analysis [5], [45], [47], [48]. We present the distribution of 3-ES in the dataset exchanges in Figure 2.

[1] https://github.com/Dregdael/EmpatheticExchanges-dataset

B. ANNOTATION PROCEDURE

The annotation process consists of providing empathy scores to conversations in ED. That is, the annotators' task is to judge the empathy displayed by the listener role in all the ED samples. We use the 5-ES metric for our annotators, then reduce the scores to obtain the 3-ES.

We recruited annotators from our institution. All participants shared the following demographic characteristics: Mexican university student, in the 19-30 age group, and presented as a minimum requirement an upper-intermediate English level. Additionally, we used a psychometric test to ensure that no annotator presented a condition that would impair their judgment of empathy. The test we applied was the Empathy Quotient (EQ) [49], which has been shown to differentiate people who struggle with empathy. The cut-off point for empathetic ability was 30 out of a possible score of 80. Therefore, we collected the results from annotators who scored on or above 35 on the EQ.

We provided each annotator with oral and written instructions for the process. In these, we state the definition of empathy in Section I, to ensure a universal understanding of empathy. Likewise, we present the levels of empathy and their implications in a conversation. Finally, we provided examples of conversations with accurate labels selected by our team based on our definition of empathy. In this manner, we expected to ensure consistency and accuracy with our labels.

To create a balanced dataset, we collected subsets of 400 conversations that followed the same emotional context distribution as ED. Therefore, we retain the advantage of emotional diversity present in the original dataset [11], [12].

Each subset from ED was sent to one annotator in our pool ($N = 10$) who would manually label the conversations according to our 5-ES metric. Therefore, we could ensure the combined dataset was balanced even if some annotators did not complete the task. Seven of our annotators met the requirements for EQ and finished the annotation process. However, data from one of them was discarded because of inaccurate labeling following manual review. In total, we obtained 2400 labeled conversations. 4 female and 2 male annotators provided the final database. We do not consider this difference in gender a cause of concern regarding accuracy, as psychological literature supports women having higher empathetic abilities than men [49], [50].

C. EXCHANGES

Another key element in creating EmpatheticExchanges is its exchange-based format. We decided to do this because empathy is a phenomenon between two or more people [1]. We aimed to capture the characteristics of an interaction. Therefore, we had to adapt ED from its original utterance-based structure to achieve our goal.

Each conversation in EmpatheticExchanges presents 2 to 4 exchanges. Each exchange is composed of a pair of utterances. First, the speaker role initiates the interaction

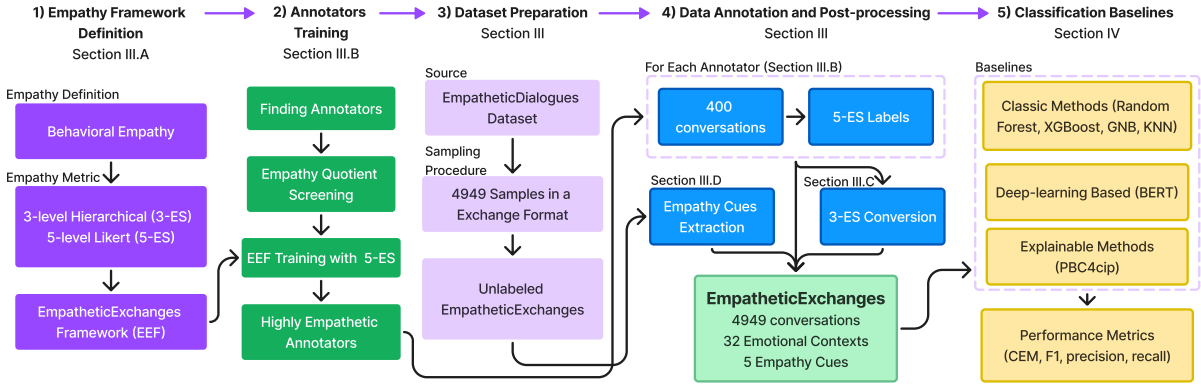


FIGURE 1. Overview of the creation process of our dataset EmpatheticExchanges and its baselines.



FIGURE 2. Distribution of 3-ES in *EmpatheticExchanges*.

by providing an utterance. Subsequently, the listener role responds to the original utterance by the previous role. This is the first exchange in a conversation. The second exchange begins when the speaker role continues the engagement by providing another utterance, followed by the listener responding again. This dynamic continues until the end of the conversation. We can see an example of a conversation in Figure 3.

Changing the format puts our dataset on par with the state-of-the-art datasets in empathy detection. Most datasets are formatted in terms of interaction. However, they present different characteristics like a focus on distressing news [6], [17], being asynchronous [1], [5], [10], and focusing on a specific domain. Our database focuses on synchronous interactions, presents a high diversity of emotion, explores different emotional situations [12], and addresses the direction of empathy [1], [10] by focusing on the responses given by the Listener.

D. EMPATHY COMPONENTS

We decided to create a database that allows researchers to study the cues and characteristics of empathy in dyadic conversations. Therefore, we need to include features beyond the text exchanges in ED. For this, we studied the nature of empathy and how elements of it can be isolated [3], [13]. This led us to the idea of separable components of empathy, which we call *empathy components*.

In addition to the dimensional model, empathy can be separated into *components* [3], [13]. These are concepts that have a connection to empathy or represent an aspect of it. Among them are cues, emotional states, feelings, or actions. They are often used in empathetic dialogue generation to improve the performance of various models [13]. In such models, the concept is translated into a measurable variable and then included in the generative process. We propose that their use in empathy detection would allow insight into the nature of empathetic communication.

Our research led us to identify a large quantity of empathetic components. Some of them are: Emotion labels [12], [20], [51], [52], [53], [54], Sentiment [21], [52], [55], Empathetic Intent [22], [23], [53], [56], [57], [58], Emotion cause [59], [60], [61], Persona [62], EPITOME communication mechanisms [20], [21], [22], [23], Dialogue strategy [63], Emotion tendency [64], and Commonsense relations [65], [66], [67]. Researchers have leveraged these empathy components to increase the perceived empathetic behavior of text generation algorithms. Because of this, we decided to adopt some of them into our database. We propose that using these components in empathy classification will increase the performance of algorithms. In addition, it will provide insight into the nature of empathy.

We identified that classification is the most popular method for obtaining components of empathy from text. The reason is that it allows researchers to include them in text generation without manual annotation. We adopt the same strategy. Thus, we use classifiers to adopt the following empathy components in our dataset: Emotion labels [11], [12], empathetic intent [57], [58], the EPITOME communication mechanisms [5], and Sentiment [21], [55], [68].

For emotion labels, we use a version of the DistilBERT model [69] fine-tuned to obtain the 32 emotional context labels from ED [70]. These 32 labels are those categories found in the original ED dataset. We provide labels for these emotions per speaker and listener utterance. Additionally, we add reduced labels based on Plutchik's wheel of

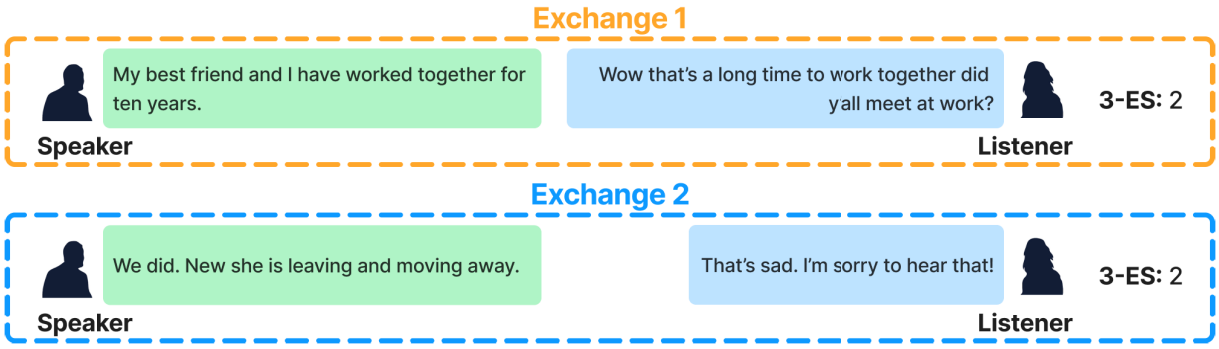


FIGURE 3. Example of a conversation in *EmpatheticExchanges*. We include the 3-ES for both exchanges.

emotions [71], [72]. We reduce to 20 and 8 emotion labels by converting the emotional context labels to their emotion equivalents in Plutchik's model [11].

Regarding empathetic intent, we use a BERT classifier [15] fine-tuned on a dataset created by Welivita and Pu [73]. This model was first proposed by Chen et al. [57]. It allows us to label our database with 8 empathetic intents and a neutral category. It is important to note empathetic intents are only from the listener role, as they represent the intent in their response to a prompt by the speaker role. We add to each data point a value for the probability that the response provided by the listener had one of the following intents: agreeing, acknowledging, encouraging, consoling, sympathizing, suggesting, questioning, wishing, and neutral.

We adopt the same model presented by Sharma et al. [5] to label our database with values for the EPITOME communication mechanisms. We use a multi-task bi-encoder model based on RoBERTa [74], which we train using Sharma et al.'s Reddit dataset. We include the score for Emotional Reactions, Interpretations, and Explorations of the EPITOME network. These scores range from 0 to 2 and represent how much a response expresses each communication mechanism.

Sentiment was obtained using a RoBERTa-base model trained on around 58M tweets and finetuned for sentiment analysis with the TweetEval benchmark [68]. This model allowed us to label the utterance for the listener and the speaker role with negative, neutral, and positive values that describe the polarity of the exchange. Similarly to empathetic intent, each data point contains positive, negative, or neutral variables that indicate the probability that an utterance presents that sentiment. We provide these variables for the speaker and listener utterances in each exchange, with an indicator "s" for the speaker, or "l" for the listener.

We identified additional empathy components that could be useful for our dataset. For example, Liu et al. [75] leverage knowledge from the NRC_VAD Lexicon [39]. In this manner, they obtain fine-grained emotional information. We adopt a similar strategy and label our database with the average values for Valence, Arousal, and Dominance (VAD) vectors for the speaker and listener roles. As such, we expected to capture the emotional dynamic per exchange. We add valence, arousal, and dominance vectors to each data point with values from -1 to 1. We provide these vectors for speaker and listener utterances, marked by an indicator at the end with the utterance role. We selected VAD vectors as part of our database because dimensional models of emotion are more flexible and capture more emotional nuance than categorical approaches, like emotion labeling [11], [76]. We were interested in its effect compared to emotional categories.

The NRC_VAD Lexicon allowed us to gauge whether mimicry was being displayed. We determined that mimicry was present in the conversation if the cosine similarity between the arousal and valence vectors from both roles was more than 0.7. We label mimicry as a binary value in our dataset.

The final list of empathy components included in our dataset is the following: VAD vectors, EPITOME mechanisms, mimicry, empathetic intent, sentiment, and emotional labels. In addition, we include two features extracted from the text: the length of both utterances in words and the exchange position in a conversation. These are not components directly related to empathy, but previous research used them [11]. We were interested in observing their effects as empathy cues on classification.

The advantage of empathy components in affective computing is that they are more easily integrated into machine learning applications. We believe components are useful indicators for identifying empathy since previous research has shown them to affect its perception levels in text generation [13]. Therefore, we propose that adding these features to our database will facilitate the empathy detection task for researchers who use EmpatheticExchanges. In particular, we expect using empathy components will improve the performance of explainable classifiers.

IV. EMPATHY CLASSIFICATION

We provide benchmarks for empathy classification in dyadic conversations using our dataset. To do this, we present a series of classical machine learning approaches, such as Random Forest, K-nearest neighbors (KNN), XGBoost [77],

and a Gaussian Naive Bayes (GNB) classifier. Likewise, we use an interpretable classifier (PBC4cip) [14]. Finally, we benchmark these approaches against a state-of-the-art BERT classification algorithm [15]. We focus on results for empathy classification using the 3-ES metric since it is more robust as mentioned in Section III.

A. EXPERIMENT DESIGN

Regarding our experimental design, we focus on two tasks. First, we aim to obtain a **classification benchmark** for our dataset. For this, we must compare the performance of various algorithms. Second, we intended to gather insight and measure the effects of individual **empathy components** in empathy classification. We tackled both tasks using the same principle. We record the performance of our classifier using different combinations of empathy components in our dataset.

B. EXPERIMENTAL PARAMETERS

Our experiments used the same 20% of instances as a **test set**. However, most algorithms used 80% of the dataset as a **training set**, whereas BERT used 64% as a training set and 16% as a **validation set**. This configuration was chosen since BERT requires a validation set. A **sample** consists of an exchange between a speaker and a listener. Our test set contains 990 exchanges, while our training set for all classifiers except BERT contained 3958. These sets can be found in our repository and were obtained through shuffling data and random sorting using the sci-kit learn library. To acquire the validation set for BERT and keep the same test set, we applied another split to the training set, which yielded training and validation sets of 3166 and 792 samples respectively.

C. EVALUATION

We carried out a comparative experiment between text classifiers. We use empathy components as our input features for all classical machine learning algorithms, as well as PBC4cip. For our results, we provide the performance of the classifiers using all cues in our database, and the best-performing combination. Meanwhile, we fine-tuned a BERT classification model [15] using the text exchanges in the dataset.

The next series of experiments explored the effects of different empathy components in classification. We used PBC4cip for this. We present its classification process in Figure 4. The experiments consisted of modifying the configuration of inputs for the classifier.

Firstly, we explored the influence of individual empathy components on a baseline. Subsequently, we performed a search strategy by adding the features in our dataset in succession to gauge their joint contribution to empathy classification. First, features were randomized in order. Then, each feature was added to the model, and the classification performance was measured. If the model increased in performance, the feature was kept. However, if the feature decreased model performance, the feature was removed. For this series of experiments, we used a 80/20 train/test dataset split. We report our findings in Section V.

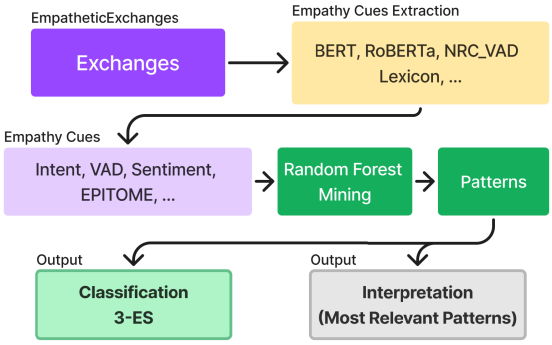


FIGURE 4. PBC4cip empathy classification process.

We used the weighted F1 score and the Closeness Evaluation Measure (CEM) [78] for our **performance metrics**. The CEM is a metric designed for ordinal classification based on *informational closeness*. It has been used for NLP tasks and empathy classification [11]. We used the CEM since our empathy labels are ordered. In conjunction, both metrics give us an overview of classifier performance on the dataset. Additionally, we provide the weighted precision and recall scores.

These experiments allowed us to provide a benchmark for empathy classification alongside EmpatheticExchanges. Likewise, they supplied information on empathy components used in affective computing and their influence on perceived empathy.

V. RESULTS

In this section, we present the results of our experimentation. Firstly, we describe the outcome of our comparison experiment. Next, we explore the effects of using different empathy components with PBC4cip.

A. COMPARISON RESULTS

We compare the performance of a state-of-the-art BERT classifier with PBC4cip and other classical machine-learning approaches. We fine-tuned BERT on the text from EmpatheticExchanges. Meanwhile, PBC4cip and other classifiers were trained using the database's empathetic cues (components). We report the results using all components and the following combination: Intent, sentiment, EPITOME emotional reactions, VAD vectors, the position of exchanges in a conversation, and the length of both utterances. This latter combination proposed yielded the best classification performance using PBC4cip. Results can be found in Table 2.

We see comparable results for PBC4cip and BERT classifiers using the F1 score and CEM when classifying 3-ES. This is observed with a similar performance when using all components and a better performance when using its best combination. Additionally, we observe that PBC4cip beats

TABLE 2. 3-ES classification results.

All components				
Classifier	**CEM**	**F1 score**	**Precision**	**Recall**
BERT	**0.591**	**0.451**	0.443	0.478
PBC4cip	0.589	0.444	0.448	0.443
Random Forest	0.569	0.401	**0.514**	**0.532**
XGBoost	0.577	0.439	0.436	0.487
KNN	0.543	0.399	0.397	0.401
GNB	0.551	0.390	0.414	0.406
Best combination				
Classifier	**CEM**	**F1 score**	**Precision**	**Recall**
BERT	0.591	0.451	0.443	0.478
PBC4cip	**0.597**	**0.452**	0.456	0.450
Random Forest	0.562	0.395	**0.514**	**0.525**
XGBoost	0.580	0.439	0.432	0.489
KNN	0.551	0.403	0.404	0.405
GNB	0.546	0.384	0.409	0.400

TABLE 3. 5-ES classification results.

Classifier	CEM	F1 score	Precision	Recall
BERT	**0.534**	**0.31**	**0.308**	**0.323**
PBC4cip	0.509	0.277	0.285	0.268
Random Forest	**0.534**	0.248	0.272	0.318
XGBoost	0.515	0.259	0.268	0.278
KNN	0.495	0.244	0.248	0.250
GNB	0.505	0.246	0.265	0.257

all other classifiers in both variations of empathy component combinations.

In addition to these experiments, we compared of all classifiers using the 5-ES metric. The classification performance metrics using 5-ES are presented in Table 3.

B. INFLUENCE OF EMPATHY COMPONENTS

We conducted a study to explore the effects of the various empathy components in EmpatheticExchanges. To do this, we focus on classification using PBC4cip. This was because it presented the best performance out of all classifiers that used empathy cues as inputs using 3-ES. We focus on this metric for the following experiments.

We present a baseline for our classification task using the length of both utterances in an exchange. We chose this feature as our baseline as it is not directly related to empathy in the literature and is a characteristic of the exchange. Subsequently, we individually add features from the dataset using a random order to gauge their impact on the empathy classification baseline. Table 4 shows the results from this experimentation phase.

Certain empathy components significantly impacted classification performance. The most influential components in classification were Intent, the VAD vectors, and 20 Emotion labels. Meanwhile, the only empathy components that negatively impacted the baseline were mimicry and 8 emotion labels.

We were also interested in studying the performance outcome of combining multiple empathy components. To do this, we sequentially increased the number of empathy components using our search strategy. We study their interaction and effects on empathy classification, particularly the CEM score. The results from this approach are visible in Table 5.

TABLE 4. Performance Impact of individual empathy components on the baseline. L: Length.

Features	CEM	F1	Precision	Recall
Length (baseline)	0.545	0.310	0.412	0.313
L + VAD vectors	0.573	**0.426**	0.437	**0.419**
L + EPITOME	0.552	0.325	**0.447**	0.319
L + Sentiment	0.572	0.415	0.427	0.409
L + Intent	**0.577**	0.423	0.433	0.416
L + Mimicry	0.534	0.381	0.379	0.385
L + Exchange position	0.547	0.294	0.442	0.310
L + Emotion labels (32)	0.556	0.401	0.407	0.397
L + Emotion labels (20)	0.573	0.418	0.428	0.411
L + Emotion labels (8)	0.542	0.374	0.400	0.361

The EPITOME framework has been used to detect and generate empathy and it is popular in affective computing [20], [21], [22], [23]. Despite this, there was a low increase in performance when using it with the PBC4cip classifier. We were interested in this phenomenon, so we tested whether eliminating aspects of it would change the impact on the classifier. The results from these experiments led us to obtain the best-performing algorithm (0.597) by eliminating the Explorations and Interpretations variables from EPITOME. This is the algorithm that we compare to BERT.

There is a clear trend that adding more information allows the classifier to detect the empathy levels of the exchanges more accurately. However, we see an exception in the case of emotion labels. This exception holds for 32, 20, and 8 emotion labels. None of the models that include emotion labels outperform the model without them. We decided to gather more information about this phenomenon. Therefore, we implemented an alternative model with all components in the best-performing model except we replaced VAD vectors with 20 emotion labels. The results showed a CEM of 0.595. This was a comparative performance to the best model.

C. PATTERNS

One of the main advantages of pattern-based classifiers is that they mine and store patterns. These patterns can later be used to gain insight from the data [11]. In our case, we stored the patterns from the different implementations of PBC4cip to extract information about exchanges with high, medium, or low empathy. We do not report the patterns from all 20 models; each model mines ∼100000 patterns that describe the instances. Therefore, we focus on the patterns with the highest frequency per class in our best-performing model.

The pattern format in this article uses an *s* and *l* before the feature to differentiate between speaker and listener utterances. Meanwhile, the empathy component of a feature is expressed as its initial in parenthesis. We use the set notation for feature values.

TABLE 5. Classification performance from joint empathy components. VAD: VAD vectors, EX: Exchange position, E: Emotion labels.

Features	CEM	F1	Precision	Recall
Length + VAD	0.573	0.426	0.437	0.419
Length + VAD + EPITOME	0.575	0.425	0.433	0.419
Length + VAD + EPITOME + Sentiment	0.580	0.431	0.437	0.426
Length + VAD + EPITOME + Sentiment + Intent	0.583	0.436	0.437	0.437
Length + VAD + EPITOME + Sentiment + Intent + Mimicry	0.590	0.439	0.443	0.436
Length + VAD + EPITOME + Sentiment + Intent + Mimicry + EX	**0.593**	**0.449**	**0.452**	**0.448**
Length + VAD + EPITOME + Sentiment + Intent + Mimicry + EX + E (32)	0.589	0.444	0.448	0.443
Length + VAD + EPITOME + Sentiment + Intent + Mimicry + EX+ E (20)	0.589	0.441	0.443	0.443
Length + VAD + EPITOME + Sentiment + Intent + Mimicry + EX + E (8)	0.586	0.441	0.445	0.440

The following pattern:

$$P1 = \{l_length \in [1, 6], questioning(I) \in [0, 0.9], EX \in [1, 2], l_neutral(S) \in [0.2, 1], s_negative(S) \in [0, 0.1], s_length \in [1, 21], suggesting(I) \in [0, 0.005], s_arousal \in [-1, -0.1], neutral(I) \in [0, 0.09]\}$$

is the one with the highest frequency for the *little to no empathy* class. The most frequent pattern in the *somewhat empathetic* class is the following:

$$P2 = \{l_negative(S) \in [0.01, 1], l_length \in [1, 6], s_negative(S) \in [0.1, 0.8], wishing(I) \in [0, 1], questioning(I) \in [0.005, 1], l_neutral(S) \in [0.06, 0.8], l_arousal \in [-1, 0.08], l_valence \in [-1, 0.5]\}$$

Meanwhile, the pattern most frequent for instances in the *empathetic* class can be found below:

$$P3 = \{l_length \in [13, 88], l_negative \in [0, 0.01], s_positive \in [0, 0.9], acknowledging(I) \in [0, 0.9], s_valence[-0.1, 1], ER\ (EPITOME) \in [1, 2], neutral(I) \in [0, 1], encouraging(I) = 0, s_arousal \in [-1, 0.06], agreeing(I) \in [0, 0.9], suggesting \in [0, 1], l_dominance \in [-0.05, 1]\}$$

These patterns present large quantities of features, which makes them difficult to interpret. However, they yield information about the nature of the conversation. In the case of the best-performing model, these patterns are unique to the class they most support. However, there are models where the pattern with the most support is the same for multiple labels.

We include the patterns for all 20 models in our public repository. In addition, we provide a document that contains the patterns with the most support.

VI. DISCUSSION

The contributions of this article concentrate on a novel dataset, EmpatheticExchanges, for the task of behavioral empathy classification. In this section, we provide further insight into the classification results of all of our proposed empathy classifiers, which serve as a baseline.

A. CLASSIFICATION

The classification comparison experiment shows that our dataset allows explainable approaches to perform similarly to deep learning ones using our 3-ES metric. BERT and PBC4cip present the same metric under the F1 score: 0.45. This metric is robust for NLP classification tasks. However, it is not suitable for ordinal classification [78]. Hence, we focus our attention on the CEM results. Both classification algorithms outperform all other machine learning approaches.

CEM displays the performance of all classifiers more accurately since it is designed for ordinal problems. The results using this metric show an increase in the performance of PBC4cip using a specific combination of features compared to the BERT classifier. This supports the hypothesis that both classification approaches are of comparable performance. This finding is valuable since the contrast pattern-based approach uses an abstraction of empathy. The results imply that using this abstraction can be sufficient to capture empathy in interactions between two people. Therefore, this strategy could be applied to other types of interactions in another medium using the same empathy components.

Another way to examine a pattern-based approach and the BERT classifier is that they are bottom-up and top-down approaches, respectively. The top-down approach (BERT) allows the classifier to gauge the nature of empathy through learned parameters. In contrast, we built a representation of text exchanges using empathy components in the bottom-up approach. Representing empathy as a series of features and traits has been observed in psychological research [3], [19], [79], [80]. We mimic this method for text interactions. Subsequently, our results support the validity of a bottom-up strategy since it demonstrates the value of our database features in this manner, as they are useful for empathy detection. Furthermore, our novel representation approach in our dataset can be used in future research to better understand its effects on empathy in different contexts besides the conversations in our dataset.

Our classification experiments focus on the 3-ES metric since it is more appropriate for measuring empathy than 5-ES. This is due to two factors: unclear empathy levels, and psychological support. We observed the first factor during annotation, in which participants were unsure of the difference between some levels of empathy. Likewise, the second factor refers to the fact that 3-ES is based on

the hierarchical nature of empathy. Using the 5-ES metric, we noticed that all classifiers lower their performance, which can be attributed to the disadvantages of the metric and a possible noise increase. Nevertheless, our 5-ES experiment also contributes to the hypothesis that using empathy components can match the performance of deep-learning methods (BERT), as Random Forest and XGBoost achieved comparable performances.

B. EMPATHY COMPONENTS AND THEIR INFLUENCE

One of our findings is that there is a difference between using single and combined empathy components. The performance of single components is lower than when combined with others. Furthermore, combining components increases performance even with those that have a low or negative impact. Psychological research states that empathy components interact and build upon one another [3] and missing certain components can decrease perceived empathy in communications [79]. Our results support these statements in textual communication. Therefore, we can state that including more empathy components is a sound strategy for increasing perceived empathy. However, we found counterintuitive results when using mimicry and emotion labels.

Mimicry is one of the most peculiar cases in our experiments. It is noted as the lowest form of empathy (affective dimension) [46], [48]. We expected it to be significant in detecting empathy. However, we see that only using mimicry negatively impacts the perception of empathy by our classifier. Despite this, it does contribute to an increase in performance when combined with other components. These results support empathy being different than emotional mimicry and show evidence that the other empathy dimensions are necessary to represent this concept.

Emotions are also an important concept related to empathy through its affective dimension. In our work, we explored the effect of categorical and dimensional representations of emotions. Emotion labels and VAD vectors improved the classification compared to our baseline, as was expected. However, these emotional representations showed interference when combined. We can state that this is the case since both features present similar results when isolated. This result highlights the incompatibility of different emotion representations [37] but does not negate the importance of emotions in empathy detection. It is interesting to note that VAD vectors present an effect in performance equal to that of an expanded emotion model beyond Plutchik's 8 main emotions [71], [72], this could indicate that categorical models of emotions must be larger to reach the same flexibility and nuance of dimensional models. Nevertheless, we see that making the categorical model too large causes a decrease in performance. Future research could focus on exploring this in more detail.

Intent and the EPITOME framework are popular empathy components in affective computing [20], [21], [22], [23], [56], [57]. As such, their positive effect was expected. However, we observed a low impact from the EPITOME framework and a performance increase when most elements were eliminated. Therefore, our research calls into question the applicability of EPITOME outside the mental health support context. Intent does support the perception of empathy as expected, which leads us to postulate that it is a preferential empathy component over EPITOME in affective computing. This is also supported by the similarity between *explorations* and a *questioning* intent and between *interpretations* and the *acknowledging/agreeing* intents. Using both components might be redundant.

Finally, positive and negative empathy present different dynamics between valence in communication [19]. It is, therefore, necessary to consider sentiment to capture this nuance. Our research supports this assertion as this component is relevant in both experiments.

C. PATTERN INTERPRETATION

PBC4cip mimics the perception of humans as it is aware of certain characteristics of empathy communication and arranges them in patterns, a capability integral for the human brain [81]. We combine this alongside its interpretability to obtain more insight into empathy cues in dyadic conversations. In this manner, we explore the component's dissociability and the combined effect on everyday interaction.

Patterns P1, P2, and P3 (presented in Section V-C) provide information on empathetic behavior and how empathy components contribute to it. For example, we can interpret the pattern P1, the representative of class 1 (little to no empathy), as ''short questions with neutral polarity to longer and low-arousal prompts are unempathetic.'' This seemingly conflicts with psychological literature that presents asking questions as a fundamental process for active listening [80], [82] and being perceived as empathetic [79]. However, when we analyze the conversations we find that these patterns include exchanges like ''S: I was upset to see someone put a dent in my door. L: Car door?'' in which questions are not engaging in the affective state of the speaker role and are not open-ended. This interpretation allows us to discuss and present evidence for the nuance of displaying elements of empathy, as they are not always a sure way to increase perceived empathy.

P2 and P3 are interpretable in a manner that supports the state-of-the-art in psychological research. P2, the most frequent pattern in class 2 (somewhat empathetic), describes exchanges in which negative utterances by a speaker are responded to with a negative (mimicry) or neutral sentiment with a well-wishing (e.g. *good luck*) or questioning intent. This tracks with the idea that the listener cares about the speaker's well-being [7], [79]. However, the low arousal and negative to somewhat positive valence might be perceived as boredom [83] or lack of genuine interest. Therefore, the pattern represents the class *somewhat empathetic*.

The pattern P3 is the most frequent in class 3 (empathetic). It is a clear example of attributes from empathetic

communication such as experiencing an emotional reaction, agreeing, acknowledging, and suggesting positively in response to calm and relaxed utterances. We can interpret these responses as an attempt to build rapport. Additionally, we see a high dominance which denotes confidence, a largely understudied factor in empathetic communication.

Finally, we can argue based on our results that artificial empathy requires more than emotional mimicry. The affective dimension is quite important for its display. However, it is not sufficient. We see that empathy is conformed from a series of components that interact with each other. For empathetic behavior, it is necessary to acknowledge specific intents and factors like sentiment and even the length of utterances. We cannot state that these components are the only ones that comprise empathy. Empathy is a nuanced subject and we expect future work in affective computing to model it as a multi-component approach.

VII. CONCLUSION

In this article, we presented a novel dataset and research task. Our dataset, EmpatheticExchanges, contains 5k exchanges annotated with two metrics of empathy supported by psychological research. In addition, we introduced a series of features from state-of-the-art affective computing. These characteristics combine with the ones inherited from the EmpatheticDialogues to create an emotionally balanced dataset for detecting empathy in dyadic text conversations.

Subsequently, we provided a benchmark for classification using this database. We achieved a CEM score of 0.597, compared to a deep-learning approach of 0.591 using our 3-ES metric. This benchmark is achieved by an XAI classifier that uses patterns and explores the merits of popular features in affective computing for empathy classification. Our findings call into question using features like the EPITOME framework in day-to-day emotional situations. Furthermore, we interpreted some patterns from our research and reported that our model correctly captures the characteristics of empathy as described by psychological literature. Moreover, we provided additional observations that could be further explored in that field. These results show that explainable AI can greatly impact the field of affective computing. We intend to conduct future work to improve classification benchmarks using additional empathy cues and explainable classifiers.

We expect that our novel dataset propels the field of affective computing and artificial empathy. Particularly, we anticipate that future researchers will use our dataset and our benchmarks to carry out development in empathy detection. Likewise, we contribute with insights on the components and cues of empathy in interactions between people that we expect will be useful for future studies.

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